Mukundh Jayapal

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Summary

Applied AI engineer who builds GenAI systems and the evaluations that show whether they work, across multimodal pipelines, fine-tuning datasets, LLM-as-judge evaluation and RAG. Seeking full-time AI engineering and data science roles from Jan. 2027.

Education

McGill University — Montreal, Canada
M.Sc. in Computer Science (Non-Thesis) | *GPA: 3.83/4.0* — *Expected Dec. 2026*

- Graduate Teaching Assistant, School of Computer Science (Fall 2025 – present): COMP 445, 189, 206, 202 and 370/570.
- Mila – Quebec AI Institute, Venture Scientist Bootcamp (Fall 2025): pitched Lympheo, automated lymphedema tracking.

Vellore Institute of Technology — Chennai, India
B.Tech. in Computer Science and Engineering (AI/ML) | *GPA: 9.06/10.00* — *2025*

Experience

Trebble FM — May 2026 – Sep. 2026
AI Engineering Intern (Mitacs) — *Montreal, Canada*

- Designed a multimodal LLM pipeline that extracts editing decisions from edited videos, defining the edit taxonomy, prototyping per-type detectors (Gemini, OCR) and grounding detection in TransNetV2 shots and Whisper word timings.
- Co-developed and hardened the production pipeline for unattended Azure OpenAI runs, with GPT-5.5 shot labelling, tiered model routing for action mapping, automated quality gates, and resumable batches with model fallbacks.
- Fine-tuned GPT-4.1 and Qwen3-32B for B-roll suggestions and redesigned the editing model's training targets from transcript rewrites (about 85% copied text) to short quote-anchored JSON edits, adopted for the team's Gemma 4 31B fine-tune.
- Built an LLM-as-judge evaluation, validated with planted errors, showing frontier models made better edits and the Gemma fine-tunes (SFT, DPO) won only on restraint and cost, so the team moved from retraining to data quality.
- Audited and shipped the final 15K-example dataset for video edit recommendation (87K edits across 27 product actions), with every edit pinned to an exact transcript position, 99.7% with a written rationale and leak-free splits.

Defect Scanner — Summer 2024, Jan. 2025 – Jun. 2025
AI Engineer Intern — *Chennai, India*

- Deployed and supported real-time computer-vision SOP-adherence systems for clients including Hyundai, Maruti Suzuki and Schneider Electric, triggering conveyor stops and relay-driven tower lamps on violations.
- Prototyped an LLM-powered interface that turns plant operators' natural-language requests into code changes, auto-committed to GitHub and rolled out to edge servers with sub-second restarts (kill signals, dependency restarts).
- Built detection and segmentation datasets and trained YOLO models for 5+ projects, including per-variant underbody sealant verification and missing-part detection under variable lighting.

Bin Sari Specialized Technologies — Sep. 2023 – Nov. 2023
Software Trainee Engineer — *Dubai, UAE*

- Designed an IoT smart-parking system with license-plate recognition and a database backend for real-time monitoring.
- Built an ML corrosion-detection system for oil and gas pipelines and presented AI solutions at ADIPEC 2023.

Projects

ForecastBench Expert-Persona Analysis | *Python, LLM evaluation*
github.com/AndrewRWilliams/ForecastBench_expert_personas

- Engineered an evaluation pipeline stress-testing expert personas under adversarial or sycophantic user pushback.
- Designed LLM judges that quantify reasoning drift, showing expert personas drifted less than baseline models.
- Ran the final evaluation on 9 models: topic-specific expert personas improved Brier scores for 7 of them.

Visual RAG Financial Analyst | *LlamaIndex, YOLOv8, GPT-4o Vision*
github.com/Mukundh0007/visual-financial-rag

- Built a multimodal RAG system for 10-K filings: YOLOv8 detects table layouts and GPT-4o Vision reads what OCR misreads.
- Parallelized PDF ingestion and vectorization, cutting document processing time by 80%.
- Implemented image-level citations for every answer, giving each figure an audit trail back to the source filing.

Skills

Languages: Python, SQL, R, C++, Java, Bash
GenAI & LLMs: Azure OpenAI, Gemini, Together AI, OpenRouter, LlamaIndex, Hugging Face, Langfuse
ML, Data & Vision: PyTorch, TensorFlow, scikit-learn, YOLO, OpenCV, CVAT, Roboflow, Pandas, NumPy, Tableau
Cloud & Edge: Azure (AI Foundry, Blob Storage), GCP, AWS S3, NVIDIA Jetson, Raspberry Pi
Developer Tools: Git, Docker, Linux, Claude Code, Codex, Cursor, GitHub Copilot
Publications: SwinPyramidNet (Frontiers in Signal Processing, 2026); Corrosion Detection Using YOLO (IEEE MoSICom 2024)
Spoken Languages: English, Hindi, Tamil (fluent); French (beginner)